

Progress Report: RS-GNN, from v1 (Resonance-Symbolic) to v2 (ED02)

Temporal Interaction / Link Prediction - Duong Viet Hoang

1. Summary

Version 1 (**RS-GNN**, *Resonance-Symbolic Graph Neural Network for Temporal Interaction Prediction*) framed the problem as **temporal graph compilation**: a four-layer neuro-symbolic architecture that enforces a sequence of structural constraints end-to-end, and was sold on transductive accuracy together with interpretability, counterfactual reasoning, and robustness - largely from single-run evidence.

Version 2 (**RS-GNN**, *Regularization-by-Decoupling for Inductive Temporal Link Prediction, with a Faithful Intervene-able Lifecycle Readout*) keeps the same underlying components but **inverts the organizing principle**. Instead of coupling everything into a compiler, v2 separates the continuous representation from the link-prediction loss with a single stop-gradient, and shows - under a multi-seed, controlled protocol - that this decoupling is the right default for the **inductive** setting. The symbolic lifecycle readout is retained as an honest, scoped add-on rather than a headline capability. Several v1 claims that did not survive controlled re-testing are corrected or withdrawn here in the open, which we regard as a normal part of moving from an aspirational design to a measured one.

2. Architecture: v1 $\rightarrow$ v2

The single largest change is structural. v1 is a four-layer constraint-enforcement compiler trained end-to-end; v2 is a two-stream design in which the backbone is shaped only by a representation-level objective and the lifecycle readout sits on a stop-gradient branch.

v1 - RS-GNN: four-layer neuro-symbolic compiler

RS-GNN: Progressive Constraint Enforcement Pipeline

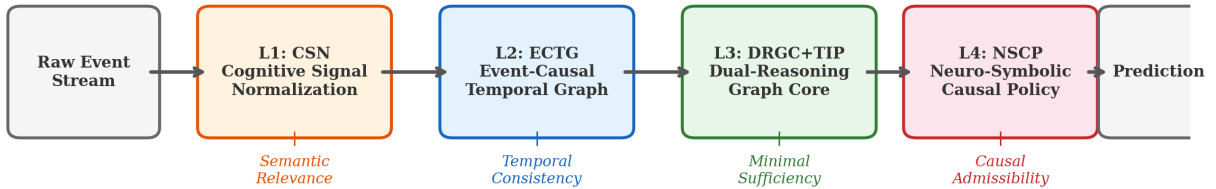


Figure 1: v1 - RS-GNN: four-layer neuro-symbolic compiler

v2 - RS-GNN: two-stream decoupled design

- **v1** chains four layers, each enforcing one constraint - semantic relevance, temporal consistency (a lifecycle finite-state machine), minimal sufficiency (a coupled recurrent

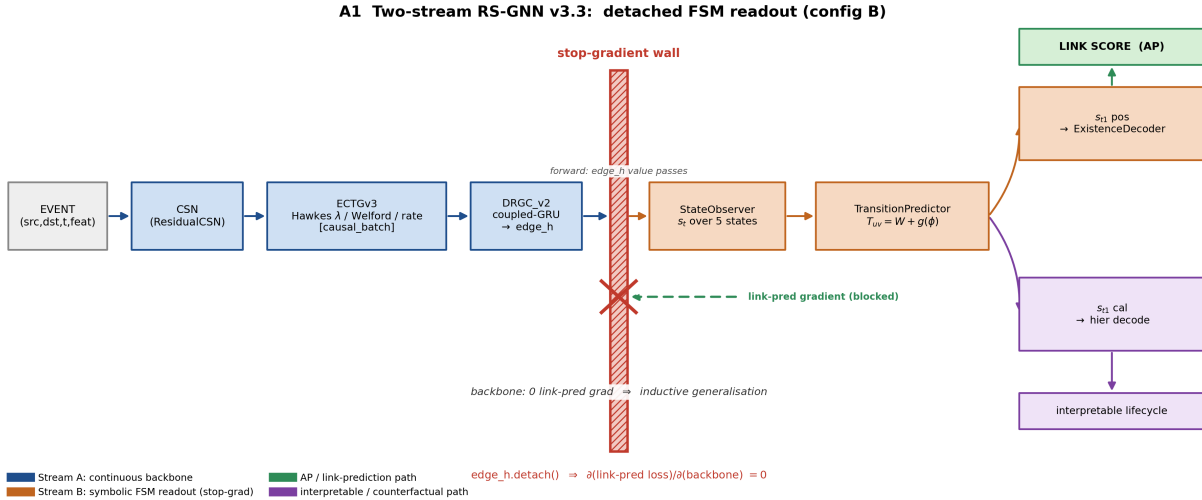


Figure 2: v2 - RS-GNN: two-stream decoupled (VAE-KL backbone + detached lifecycle readout)

memory), and causal admissibility (a hard mask on the predictions). The link loss flows through the entire pipeline.

- **v2** splits the model into two streams separated by a stop-gradient. The continuous backbone is shaped **only** by a variational / parsimony objective and deterministic update laws; no link-prediction gradient reaches it. The symbolic lifecycle readout reads from the backbone on a detached branch, so it cannot influence the scored prediction.
- The same components survive (event encoder, per-pair operator, coupled-GRU memory, five-state lifecycle machine, causal mask), but the design goal moves from “enforce constraints on the output” to “protect the representation from the link loss, and let the readout ride along for free.”

A four-panel overview of the full v1→v2 change set is below.

3. Repositioning: aspirational compiler → measured decoupling

Aspect	v1 (Resonance-Symbolic)	v2 (ED02)
Core claim	a temporal graph <i>compiler</i>	one measured design decision: decoupling
Hero axis	transductive accuracy	inductive accuracy and the gap one flag opens
Interpretability	a regulatory <i>requirement</i>	a scoped, honest add-on
Evidence	mostly single-run	multi-seed, mean ± std, uncertainty labelled

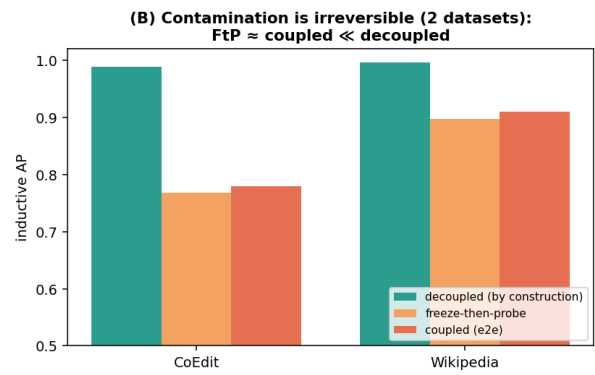
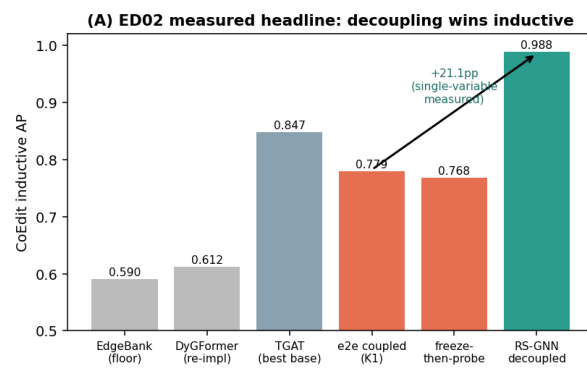
v1 sold an architecture as a normative framework supported by an optimality theorem. v2 sells a single design decision, supported by a controlled single-variable ablation and an irreversibility control, and demotes everything else - interpretability, counterfactual reasoning, the lifecycle machine - to a clearly bounded add-on. The regulatory-necessity rhetoric and the compiler metaphor are removed.

4. v1 claims corrected or withdrawn in v2

Moving to a controlled protocol surfaced several v1 claims that did not hold up. We record them here rather than quietly dropping them.

- “Resonance” / echo memory. The v1 name and story leaned on an echo-style reso-

RS-GNN: v1 (Resonance-Symbolic) → v2 ED02 : from argued compiler to measured decoupling + honest scope



(C) Retracted from v1 (integrity)

- R1 echo/resonance memory → VAE-KL regularizer
- R2 transition-CE term → de-collapse CE (detached)
- R3 regime-switch advantage → **FALSIFIED** (loses to CAWN)
- R4 causal-confidence = error flag → retracted (anti-calibrated)
- R5 do(·) = Pearl identification → typed forward re-evaluation
- Proposition 1 optimality theorem → removed
- 29.7%-causal-chain headline → removed

(D) New in ED02 (measured, 3-5 seed)

- knob ablation: enable_main_predictor → +21.1pp (1 flag)
- freeze-then-probe irreversibility → 2 datasets
- 5-seed headline → B-C +22.8 / B-TGAT +14.1
- cross-dataset knob → wiki +8.6 / mooc +0.85
- trained-0 ladder → survives training (non-tautological)
- EdgeBank floor + DyGFormer baselines (measured)
- Fig 6/7: faithful + intervene-able readout

Figure 3: Overview of the v1 → v2 changes

- **Confidence flag as a reliability signal.** v1 presented a causal-compliance score as an actionable confidence flag that alerts operators to likely errors. In v2 the successor signal reliably predicts the model’s own internal rule violations, but does **not** predict actual prediction misses (the external error-prediction result was seed-fragile and did not replicate). It is reported only as an internal self-consistency measure.
- **Counterfactual reasoning as causal identification.** v1 framed its counterfactual mechanism as Pearl-sense, do-calculus causal identification with “exact” explanations. v2 keeps the intervention operator only as a **typed input override** and explicitly disclaims causal identification: the structure is designer-imposed. The state ordering used by the operator was hand-set at initialization but is **confirmed by training** (a partial order over the lifecycle states survives optimization across all seeds).
- **Optimality theorem.** v1 stated an optimality proposition with a three-step proof claiming the compiled graph is the minimal sufficient interaction graph. v2 carries **no** optimality theorem. The single formal statement is an eval-time score-invariance property, labelled as an autograd-certified property of the trained instance, not a theorem.

Also dropped as framing (not factual corrections): the regulatory-necessity argument, a single-run “causal chain tracing” descriptive statistic, and a “only model that can do X” capability matrix - all replaced by measured deltas.

5. New measured results in v2

All headline numbers below are reported with multiple seeds and per-cell standard deviation. (Full reproducibility detail is provided in Appendix A of the paper.)

- **Single-variable decoupling ablation.** Flipping the one flag that couples the representation to the link loss - every other setting fixed - costs **−21.1 pp** inductive accuracy on CoEdit ($0.990 \rightarrow 0.779$, 3 seeds). Two adjacent gates are inert (≤ 0.3 pp), isolating the effect to this one decision.
- **Irreversibility control, two datasets.** Decoupling by construction reaches 0.988 inductive accuracy on CoEdit, whereas a classic *freeze-then-probe* on a coupled backbone recovers only **0.768** - essentially the coupled level (0.759). The same pattern holds on Wikipedia (freeze-then-probe **0.897** $\approx$ coupled 0.909 $\ll$ decoupled 0.996). The damage from coupling is therefore **irreversible**, confirmed on two datasets rather than one.
- **Five-seed headline.** On CoEdit, the decoupled configuration reaches 0.988 ± 0.003 inductive accuracy versus 0.759 ± 0.014 coupled and 0.847 ± 0.013 for a strong baseline - a **+22.8 pp** and **+14.1 pp** margin respectively.
- **Cross-dataset generalization.** The same single-flag knob, applied to two standard graphs we did not construct, yields **+8.6 pp** on Wikipedia and **+0.85 pp** on MOOC (CoEdit +21.1 pp). The magnitude tracks how much exploitable node identity each dataset contains, indicating the mechanism is not a CoEdit artifact.
- **Memorization floor and transformer frontier.** A memorization baseline reaches only ≈ 0.59 inductive on CoEdit (the win is not memorization), and a strong transformer baseline behaves as expected through the same harness (recovering on Wikipedia), confirming the evaluation pipeline is not pessimizing competitors.
- **Counterfactual battery at scale.** Typed interventions on 12,000 real CoEdit pairs across 3 seeds are 100%-directional: forcing the terminal state lowers the predicted existence in every case (mean -0.522 ± 0.001), forcing the growth states raises it in every case, and the operator is exactly reversible.

6. Methodology: v1 vs v2

Aspect	v1 (Resonance-Symbolic)	v2 (ED02)
Top-level shape	four-layer pipeline	two streams + stop-gradient
Backbone signal	link loss flows through all	only a representation
Lifecycle machine	on the prediction path	objective behind the detach; heads split
Causal mask	hard mask on predictions	soft, band-diagonal mask
Counterfactual	re-run on prediction path	typed override, score unchanged
Formal backbone	an optimality theorem	an invariance property

The v1 components survive, but the organizing principle inverts. v1: couple everything end-to-end and enforce constraints on the output. v2: decouple the representation from the link loss with one stop-gradient - the single move that generalizes inductively, and that incidentally frees the readout.

7. Scope and next steps

v2 deliberately splits the story into two contributions with different scope.

- **C1 - decoupling for inductive accuracy - general.** Shown directionally on three datasets (CoEdit +21.1, Wikipedia +8.6, MOOC +0.85), with the irreversibility control on two. This is the falsifiable, generalizable core, and makes no reference to any one dataset’s domain.
- **C2 - per-pair lifecycle readout - deliberately CoEdit-scoped.** The drivers that move a pair through its lifecycle are non-commensurable across domains (e.g. a transaction amount, a counterparty degree, and an edit rate are not the same quantity). A single global readout would have to discard the per-domain driver knowledge that makes the readout faithful, so this contribution is scoped by design rather than underdeveloped. We treat this as a deliberate general-versus-faithful trade-off: C1 is shown across datasets, C2 is kept faithful within one.

Both halves share a single mechanism: the same stop-gradient blocks the identity shortcut (C1) and frees the readout from the scored path (C2). C2 complements C1 rather than competing as an independent novelty.

Future work (not claimed as done): promote the cross-dataset knob and the full configuration comparison to five seeds; add further transformer and nearest-neighbour baselines; a per-novelty-bin identity probe; and a globalized version of the per-pair readout (currently designed but not yet built).